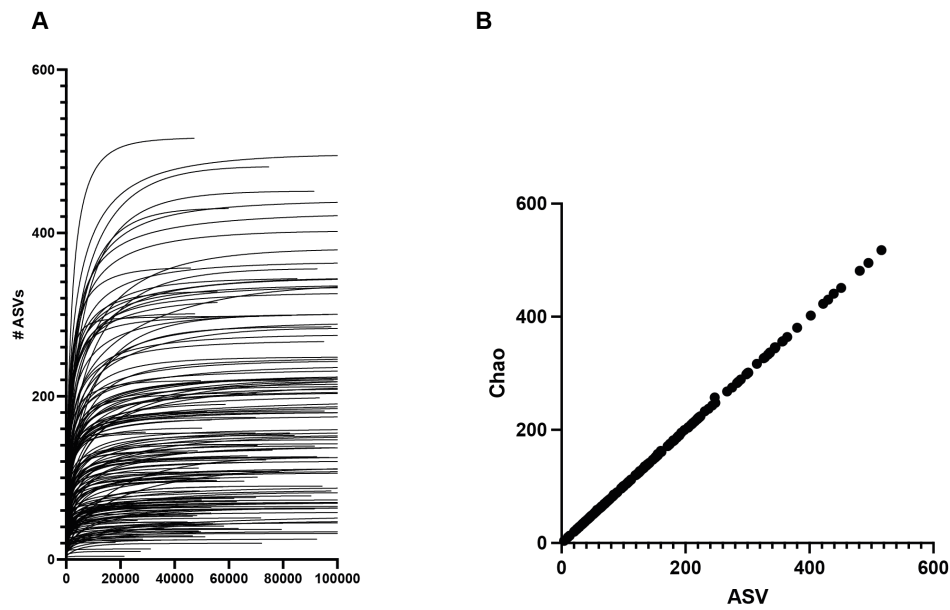
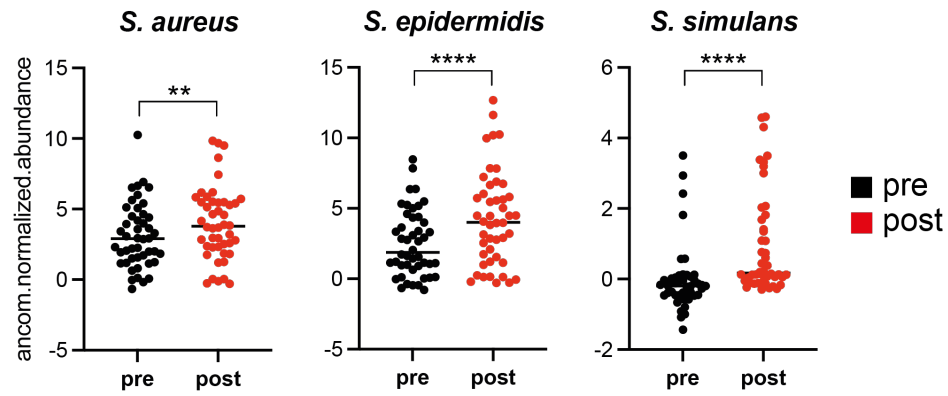
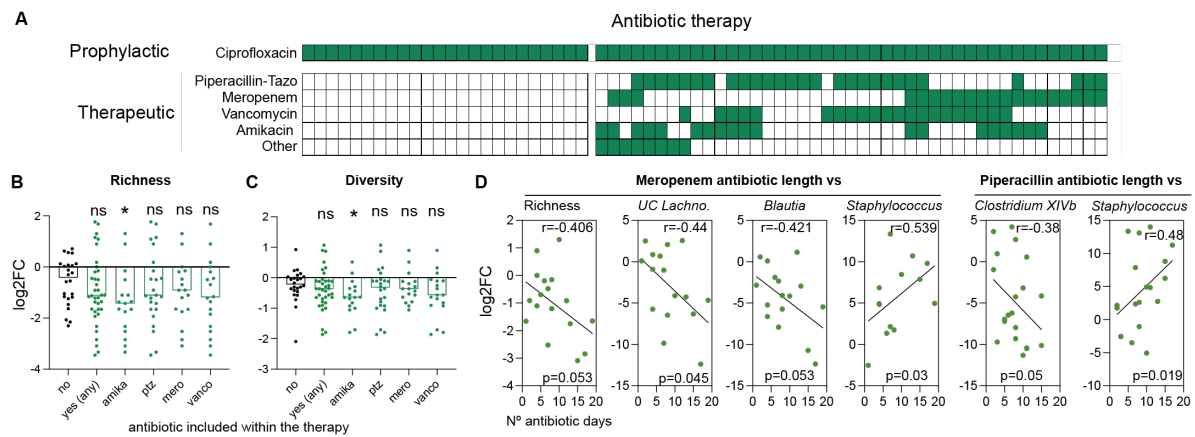




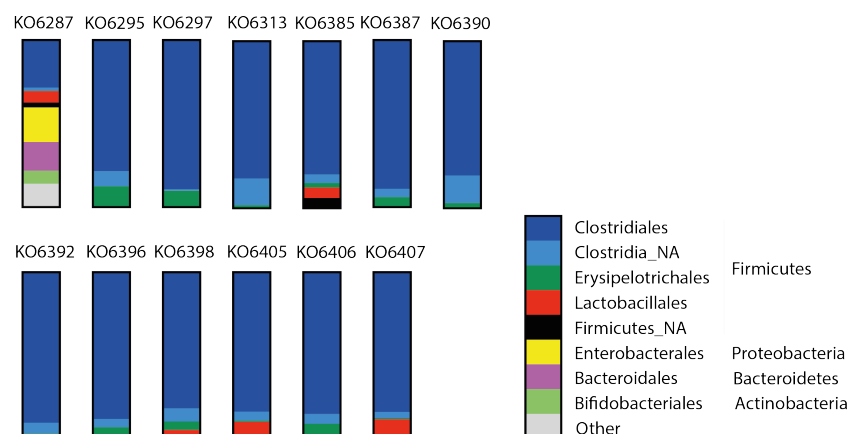
Supplementary Figure 1. 16s rRNA gene sequencing depth was sufficient to identify most ASVs in all samples. (A) Rarefaction curves representing the number of ASVs identified in each sample after analysing an increasing number of sequences. Since most samples do not contain more than 100.000 sequences, for facilitating the visualization of the figure, the ASVs identified for a maximum of 100.000 sequences are shown. **(B)** The number of ASVs is almost identical to the Chao index, representing the potential maximum number of ASVs that could be detected. N=172 samples.



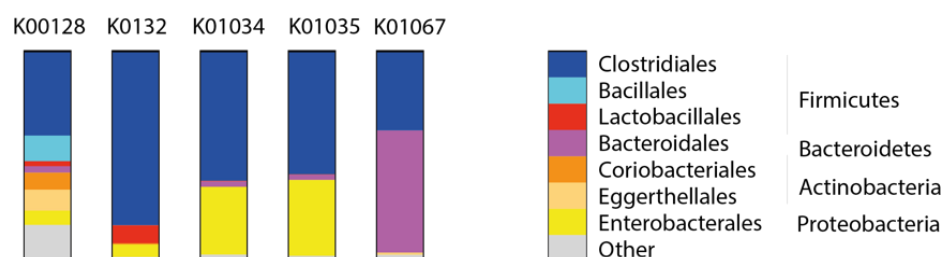
Supplementary Figure 2. Multiple *Staphylococcus* species expand during allo-HSCT. The abundance of *S. aureus*, *S. epidermidis* and *S. simulans* is significantly higher in samples collected post- allo-HSCT. ANCOM-BC & FDR. ** $p < 0.01$, **** $p < 0.0001$. N=46 patients per time point.



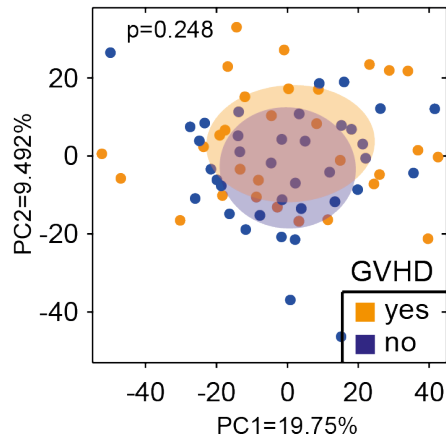
Supplementary Figure 3. Antibiotic treatment is associated with microbiota changes detected after allo-HSCT. (A) Heatmap showing the antibiotics received by different patients. Each column represents one patient and each row one antibiotic. All patients received ciprofloxacin as prophylaxis at hospital admission. A subset of patients received additional antibiotics as therapeutic agents as depicted. Other: aztreonam, ceftazalone, colistin, tigecycline, linezolid, metronidazole, teicoplanin. **(B-C)** Log2 fold change (FC) in the number of ASVs (Richness) (B) or Shannon diversity index (Diversity) (C) between the samples collected pre- and post- allo-HSCT. Patients were divided in different groups according to the administration of specific antibiotics between the 2 collected faecal samples. No: did not received any antibiotic besides prophylaxis. Yes: received antibiotics besides prophylaxis, including: Amika: amikacin, PTZ: piperacillin-tazobactam, Mero: meropenem, Vanco: vancomycin. **(D)** Spearman correlation analysis between change in richness or in the abundance of specific taxa vs days of meropenem treatment or piperacillin-tazobactam treatment. Only significant correlations that can explain the change detected in Fig. 1 are shown. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, Dunn's test for B-C and one-sided Spearman correlation test for D. Bargraphs represent the median in B-C. N= 17-37 patients for (B-C) and 17-24 for (D). Pairs of samples not containing a particular bacterium in both the pre- and post-transplant sample are not included in (D).



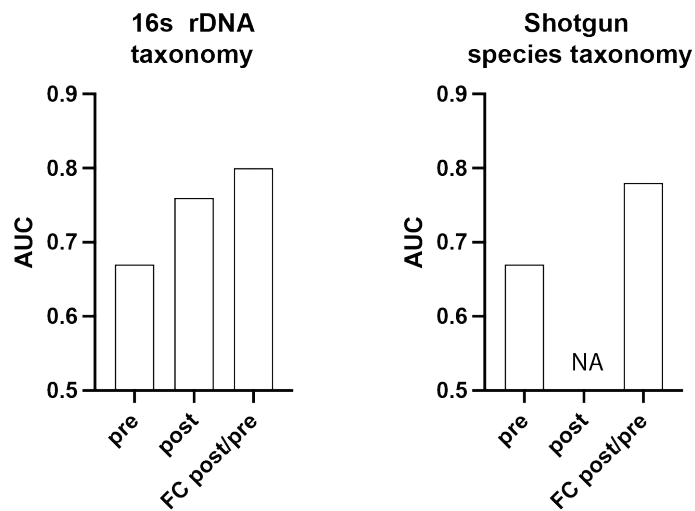
Supplementary Figure 4. Taxa encoding KOs within the “Cell growth” pathway whose abundance significantly decrease after allo-HSCT. Colours represent the % of each bacterial order encoding each KO. NA: not assigned.



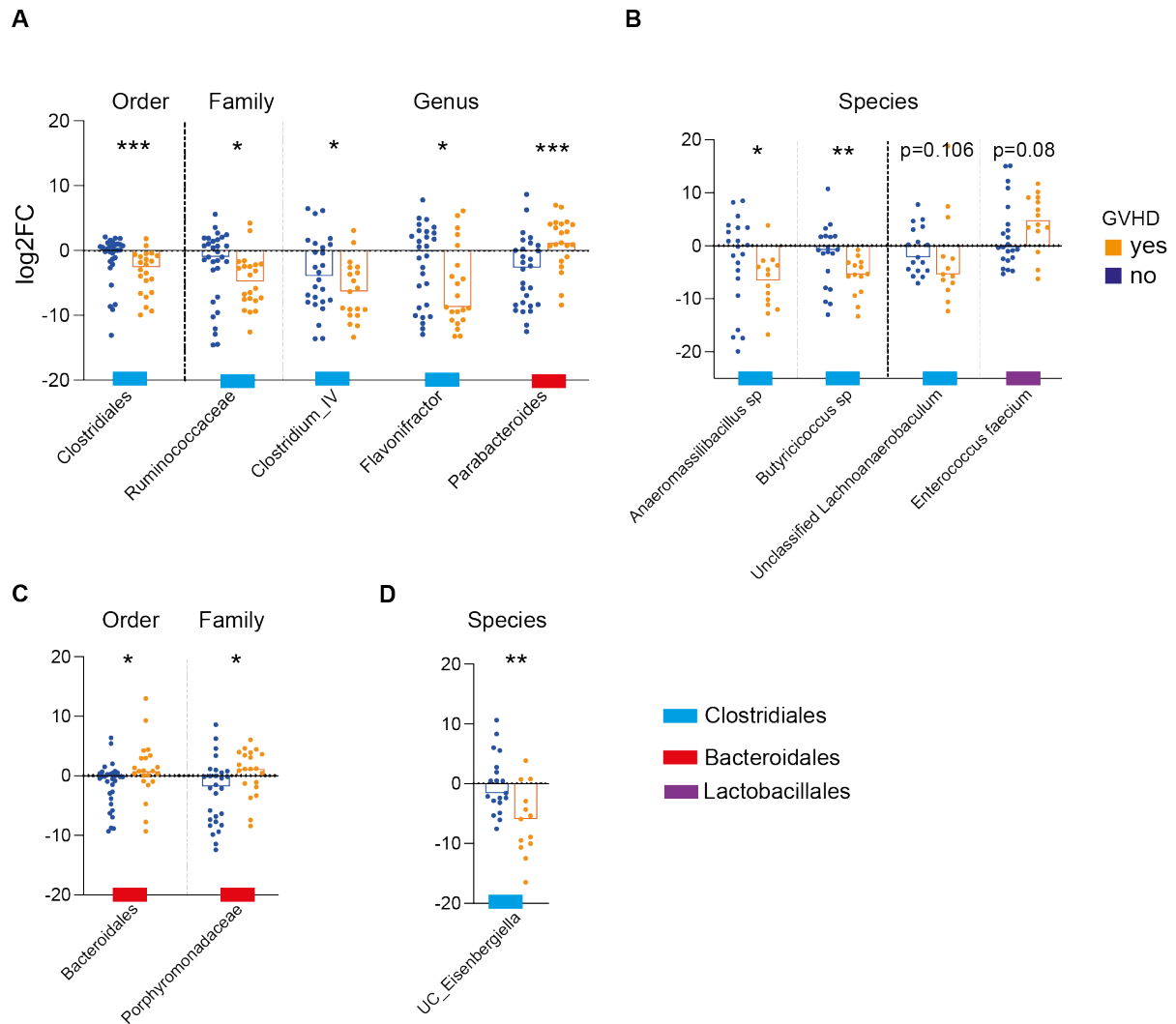
Supplementary Figure 5. Taxa encoding KOs within the final step of the pathway of synthesis of SCFAs. Colours represent the % of each bacterial order encoding each KO. Their role in SCFAs synthesis is specified in Fig. 2H and 2I.



Supplementary Figure 6. Overall microbiota changes after allo-HSCT do not differentiate patients according to their future GVHD status. PCoA analysis using Euclidean-distances obtained with a matrix containing the log2FC in taxonomic abundance between pairs of samples (post vs pre transplant). P value was calculated with PERMANOVA test. N= 37 non-GVHD and 30 GVHD patients.

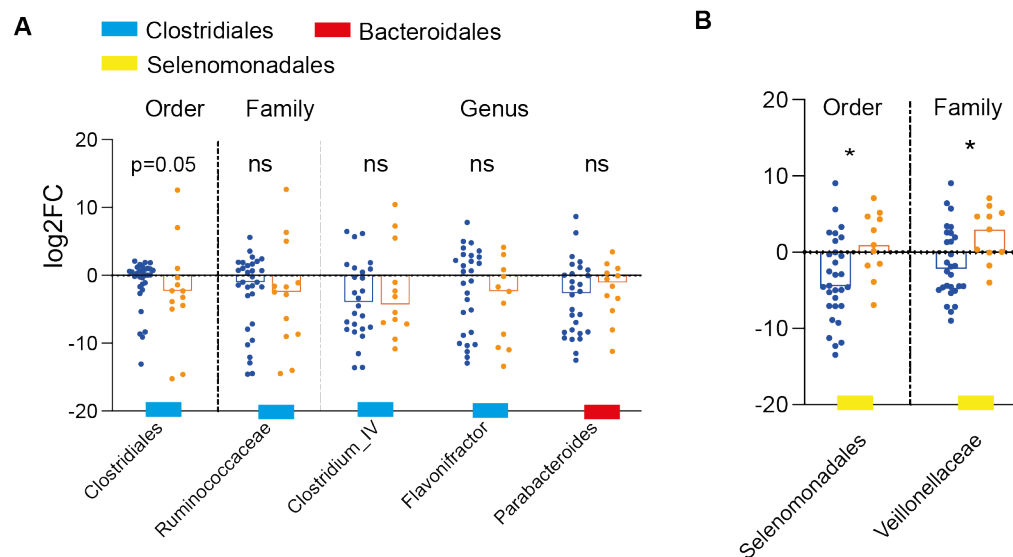


Supplementary Figure 7. Microbiota dynamics classify to a better extent patients that will develop GVHD. Discrimination power to classify patients according to their GVHD status using a random forest model built with the bacterial taxa identified by the Boruta algorithm (Suppl. Tables 7 and 8) using 16s rRNA data (class-genus levels) or shotgun sequencing data (species level). Bacterial taxa were identified either using the information obtained in samples collected pre-transplant, post-transplant or both (the log₂FC of bacterial abundance post vs pre transplant). NA: not applicable since the Boruta algorithm did not confirm any bacterial taxa with discriminatory potential at this particular taxonomic level.

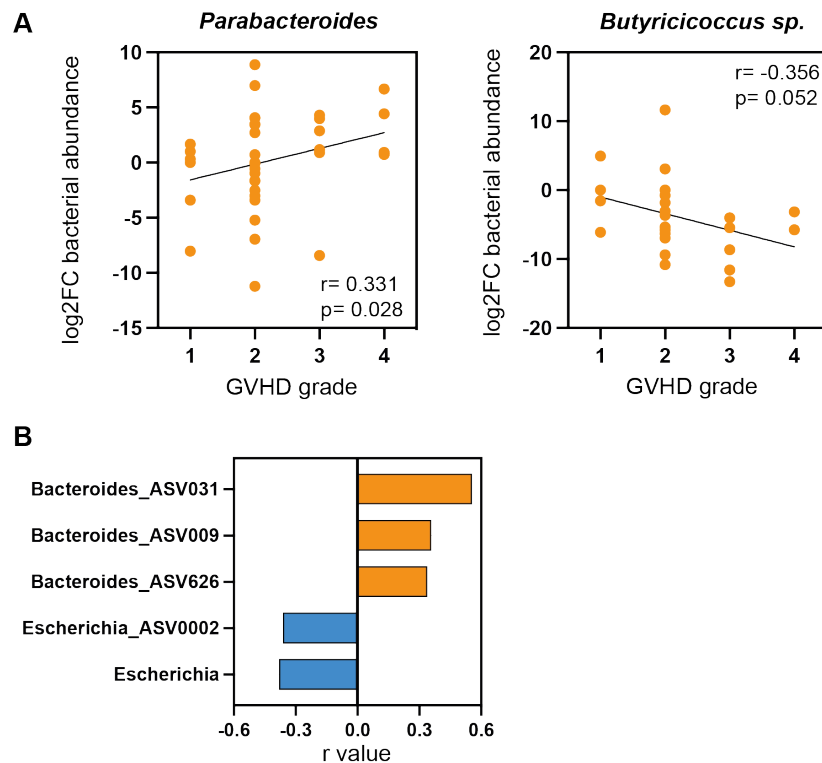


Supplementary Figure 8. Specific microbiota taxonomic changes are detected in patients that will develop GVHD in the lower gastrointestinal tract. (A-B) Comparison of specific microbiota changes between a group of patients that did not develop any sign of GVHD (no) and a group of patients that developed signs of GVHD in the lower gastrointestinal tract (yes). The studied bacteria were those found to be differentially abundant in GVHD patients in the analysis shown in Fig. 4, either using 16s derived taxonomy at the order, family and genus levels or shotgun sequencing derived taxonomy at the species level. **(C-D)** Additional microbiome changes that significantly differentiate LI-GVHD patients identified using 16s rRNA (C) or shotgun sequencing data at the species level (D). Positive numbers indicate increase after allo-HSCT, negative numbers indicate decrease. Bargraphs represent the median. Only pairs of samples with detectable bacterial levels in at least one of the samples of the pair are included in each particular taxonomic fold change representation. N= 22-31 patients (A,C) and N=14-22 (B,D). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ one-sided Mann-Whitney test (A-B) and

two-sided Mann-Whitney test (C-D). P values between 0.05 and 0.1 are indicated in the Figure. All other p values are shown as ns. UC= unclassified

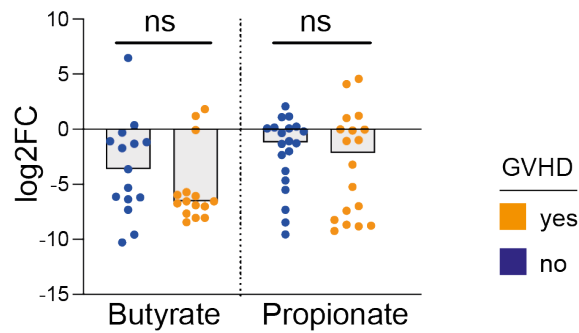


Supplementary Figure 9. Specific microbiota taxonomic changes are associated with exclusively developing GVHD in the skin. (A) Comparison of specific microbiota changes between a group of patients that did not develop any sign of GVHD (no) and a group of patients that developed GVHD exclusively in the skin (yes). The studied bacteria were those found to be differentially abundant in the analysis shown in Fig. 4. using the 16s derived taxonomy at the order, family and genus. Analysis using the shotgun sequencing data were not performed due to the limitation in the number of patients available with this data developing exclusively GVHD in the skin (N<10). **(B)** Microbiota changes that significantly differentiate those patients that will develop GVHD exclusively in the skin (Confirmed by machine learning, Boruta algorithm, and significant $p < 0.05$ in the Mann-Whitney test). Only pairs of samples with detectable bacterial levels in at least one of the samples of the pair are included in each particular taxonomic fold change representation. Bargraphs represent the median. * $p < 0.05$, ** $p < 0.01$, one sided Mann-Whitney test for A and two-sided for B. P values between 0.05 and 0.1 are indicated in the Figure. All other p values are shown as ns. N=13-31 patients.

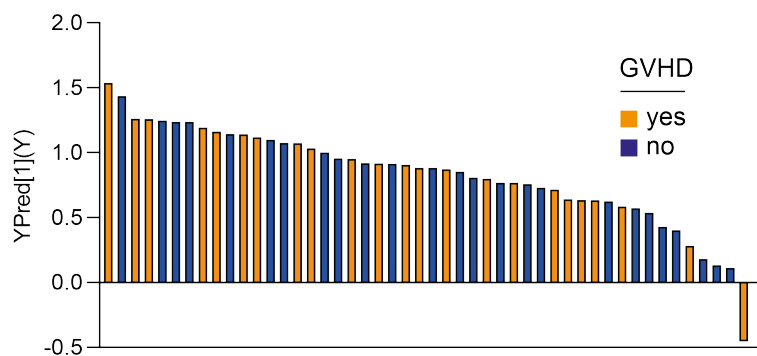


Supplementary Figure 10. Bacterial taxonomic changes correlate with GVHD severity.

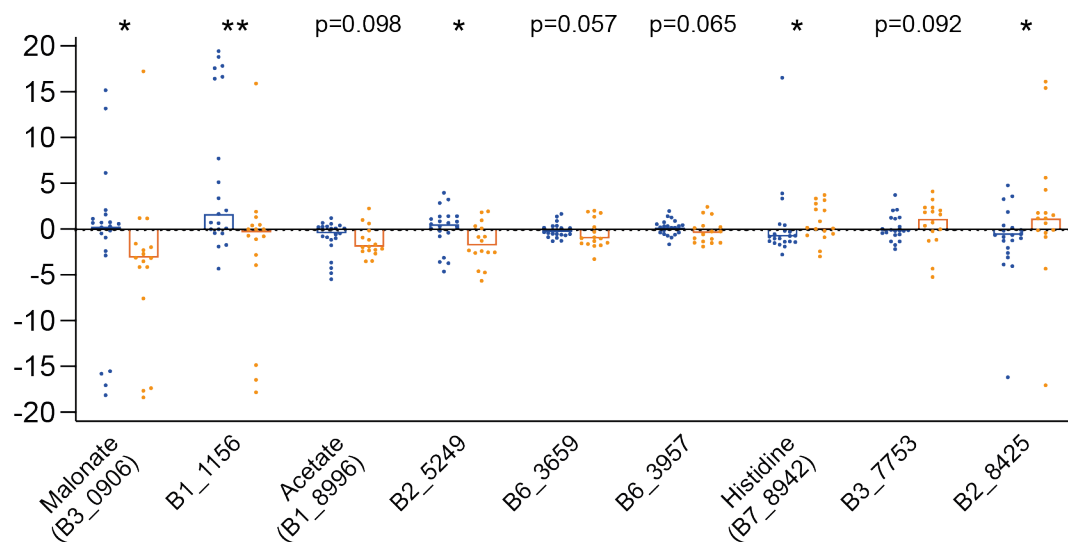
(A) Spearman correlation analysis (one-sided) between the GVHD grade and the log₂FC taxonomic abundance (pre vs post- transplant). Taxa that were studied in this analysis were those whose change over time was found to be significantly different between GVHD and non-GVHD patients. Each point represents one patient. Patients not containing a particular bacterium in both the pre- and post- transplant sample are not included. (B) Additional taxa whose change was found to be significantly associated with GVHD grade identified through PLS regression analysis ($r_{pls} > |0.4|$) and Spearman correlation test & FDR (two-sided, $q < 0.05$). Bargraphs represent the obtained Spearman correlation value. N=36 patients.



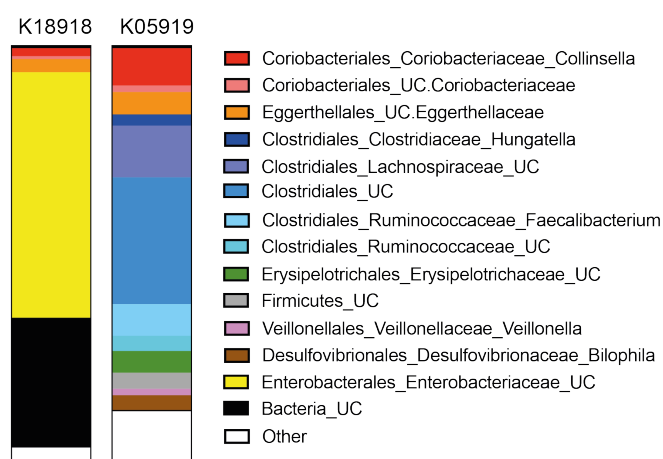
Supplementary Figure 11. Changes in butyrate and propionate after allo-HSCT according to the future GVHD status. Changes in the levels of butyrate and propionate after allo-HSCT. $^{ns}p>0.217$ Mann-Whitney test. N=15-20 patients. Only pairs of samples with detectable bacterial levels in at least one of the samples of the pair are included in each particular fold change representation.



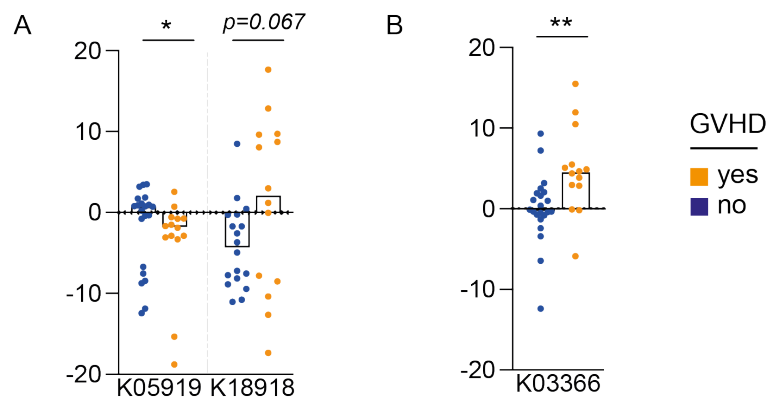
Supplementary Figure 12. Samples from GVHD and non-GVHD patients do not differ in their overall metabolome changes according to OPLS-EP analysis. Bar graph of the estimated effect (Yhat) on the metabolic profile, reflecting the difference between pre- and post-transplant over all matched sample pairs (bars in plot) coloured according to GVHD development. Yhat = 1, corresponds to the target value for the OPLS-EP model (R^2Y : 0.832, Q^2 : 0.539). The size of each bar corresponds to the dissimilarity of each pre- and post-transplant pair. Small values indicate small differences between the pre- and post-transplant samples. Large values indicate large differences, and negative values correspond to a difference in the opposite direction between matched pre- and post-transplant individuals.



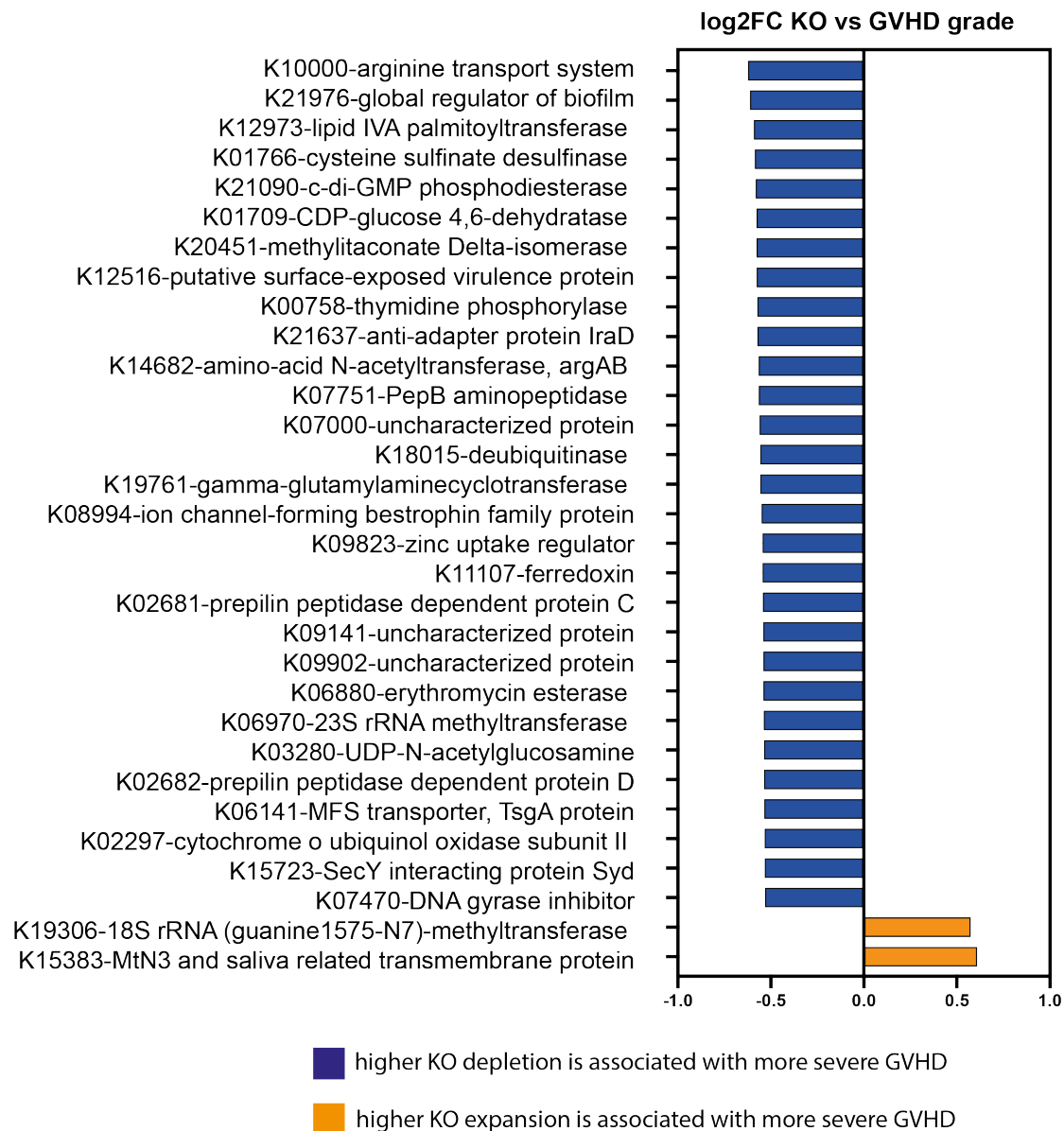
Supplementary Figure 13. Specific metabolomic changes are detected in patients that will develop GVHD in the lower gastrointestinal tract. Comparison of specific metabolomic changes between a group of patients that did not develop any sign of GVHD (no) and a group of patients that developed LGI-GVHD (yes). The studied changes were those found to be significantly different in the analysis shown in Fig. 5. Positive numbers indicate increase after allo-HSCT, negative numbers indicate decrease. Bargraphs represent the median. N= 16-21 patients. * $p < 0.05$, ** $p < 0.01$, one-sided Mann-Whitney test. P values between 0.05 and 0.1 are indicated in the figure.



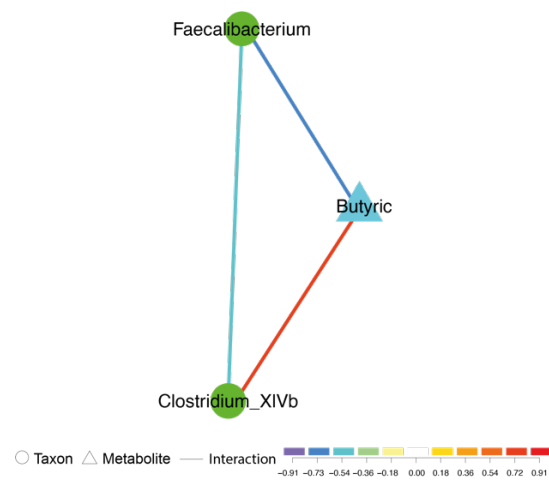
Supplementary Figure 14. Taxa encoding the K05919 and the K18918. Colours represent the relative abundance of each taxon contributing to each KO. UC= unclassified.



Supplementary Figure 15. Changes in specific KOs are associated with GVHD development in the lower gastrointestinal tract. (A) Comparison of specific KO abundance changes between a group of patients that did not develop any sign of GVHD (no) and a group of patients that developed LGI-GVHD (yes). The studied KO were those found to be significantly different in the analysis shown in Fig. 6. * $p < 0.01$, one-sided Mann-Whitney test. **(B)** KO whose change over time is significantly different between patients who will or will not develop LGI-GVHD (Confirmed by machine learning, Boruta algorithm, and significant $p < 0.05$ in the Mann-Whitney test). Only pairs of samples with detectable KO levels in at least one of the samples of the pair are included in each particular KO fold change representation in B. N=14-22 patients.



Supplementary Figure 16. Changes in specific KOs are associated with the severity of GVHD. Spearman correlation analysis between the GVHD grade and the log2FC KO abundance (post- vs pre- transplant). Bargraphs represent the spearman correlation values. Only correlations selected by a PLS-R analysis with r_{pls} values $>|0.4|$ and Spearman correlation test & FDR q values <0.05 are shown. Bargraph represent the Spearman r values.



Supplementary Figure 17. *Clostridium XIVb* is negatively associated with *Faecalibacterium* and positively associated with butyrate in post-transplant GVHD samples. Subnetwork within the network shown in Fig. 7A.